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DrukGreen

Annual Maintenance Completion Report
(2025-2026)
Generating unit-I



Mechanical Maintenance Unit
Tala Hydro Power Plant

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1. Overview

The Annual and Special Maintenance (ASM) of 6x170MW generating units and its auxiliaries are carried out annually to ensure smooth and reliable operation of the units during peak generating season. The maintenance commences from October to May for over seven months period, as the river inflow during this period decreases and the discharge is sufficient to operate less than three to four units. Opportunities are taken to attend to all major and minor maintenance works such as replacement of spares, overhauling, servicing, testing and all schedule maintenance required as per OEM on the generating units, its auxiliaries and other installation in the Power House, Butterfly Valve (BFV), steel lined Pressure Shaft and Surge Shaft during the maintenance period.

The annual maintenance work for unit No. I was completed **on 29.04.2026**. However, with the lifting of TRT gate and unit was put to mechanical spun on **15:00** hours on **30.04.2026**.

The mechanical spin was carried out on Thursday dated 30.04.2026 @ 16:30 hours and @ 18:30 hours, followed by Dry run. The scheduled and actual start and end of unit-1 AM 2025-2026 on the same day after the bearing temperature remained constant for more than 12 hours. After confirming the normalization of all operating parameters, then generating unit was shut down after dry run where LGB pad-2 temperature reached at 58 degree Celsius and further Operation Section kept observation the parameters of bearings. After dry run of 12 hours the generating unit parameters remains constant and initiated to shut down the unit. The unit was yet to initiate for connecting grid very soon.

Besides conducting scheduled maintenance as provided in the OEM manual other special maintenance were also done. The major maintenance activities conducted pertaining to Mechanical Maintenance Unit for the unit are as detailed hereunder and all test/calibration reports, checklists and inspection notes are placed at Annexures.

The Annual Maintenance of Unit-1 was carried out simultaneously with Unit-2, covering essential inspection and servicing activities to ensure reliable operation of the generating unit. The major work planned was grinding of eroded runner buckets, welding of buckets and regrinding of the welded portions and maintaining of the buckets profile. Repairing of the runner buckets was another major activity planned for the unit.

Other works carried out included pre-testing and overhauling of nozzles, replenishment of nitrogen in the MIV accumulator banks, calibration of MIV pressure gauges, cleaning of the Main Heat Exchanger, and inspection of the Cooling Water pumps. In addition, Ultrasonic Testing was conducted on the MIV decompression pipeline, bypass pipeline, and the pressure accumulators of both the MIV and governor systems.

All activities were executed in coordination with the concerned sections, ensuring adherence to safety and quality requirements. The Special and Annual maintenance Schedules are placed at Annexure 1. Scheduled vs Actual date of start and completion of maintenance on the generating unit is shown in Table-1.

SN	Unit #	Approved Schedule			Actual Completion		
		Start	End	No of Days	Start	End	No of Days
1.	Unit-I	04.03.2026	30.04.2026	26	04.03.2026	30.04.2026	26

Table 1: Unit-1 AM schedule

The major maintenance activities conducted for the unit are detailed hereunder and all test/calibration reports, checklists and inspection notes are placed as Annexure.

2. Turbine System

2.1 Nozzle Injector Assembly

All the nozzle were pre-pressure tested for its ability to retain the required amount of pressure (90-100 kg/cm²) and based on the test result; nozzle -J1 and J5 was dismantled and taken to service for overhauling.

The replacement of needle tip holding bolt M20x150mm of all the nozzles have done. One or two necessity internal fasteners were also replaced, and a balanced tightening of the fasteners was ensured. Assembling of all critical components of the nozzles was carried out under a close supervision and check formats are maintained for record and reference.

Other three nozzle injector assemblies that did not require to be dismantled except their needle tip center bolts and few numbers fasteners. All the internal fasteners were replaced with new set such as replacement of needle tip holdings bolts M20x150mm.

Table 2: Spares consumed for Turbine system

Sl . No	Materials	Qty.	Unit
1	SS Allen 130Bolt, M20x200, grade 9.8	2	NO
2	SS Stud with Nut & Spring Washer, M30x	10	NO
3	NBR Endless O-Ring 335x6mm, 70 shore	4	NO
4	Cup Sea, P.P-110X130X12mm	4	NO
5	Wiper Seal PU with casing 110x120x7/10mm	3	NO
6	Wiper Seal P.P-200x230x16mm	3	NO
7	Wiper Seal PU with casing 200x220x12/16mm	2	NO
8	PTFE Ring Seal 200x209.2x7.5mm with inner	2	NO
9	NBR Endless O-RING 246x6mm, 70 Shore	2	NO
10	HT Allen Bolt, M16x160, Grade 12.9	2	NO
11	HT Allen Bolt, M20x150, Grade 12.9	5	NO
12	NBR Endless O-Ring, 320x6mm, 70 shore	10	NO
13	NBR Endless O-Ring, 330x8mm, 70 shore	2	NO
14	NBR Endless O-Ring, 350x8mm, 70 shore	4	NO
15	NBR Endless O-Ring, 19.5x3mm, 70 shore	2	NO

Figure 1: Overhauling & servicing of nozzle injectors



2.2 Runner

The unit-I runner 5020 was in operation from 16.05.2021 and its running hours since the commission has reached 30362.41 hours by the time the unit was taken under shutdown for annual maintenance on 28.04.2026 (09:00:00 AM).

The physical condition of the runner buckets was visually inspected, and erosion was observed on all buckets. Considering the significant extent of erosion and that the runner had already operated for more than 37,149.34 hours

Table 3: Runner particulars

Particulars	Runner (5020)
Make	Andritz Hydro, Switzerland
Type	Forged Fabricated and hard coated
Hard Coating	HVOF
Runner Number	5020
Cumulative Running Hours	30362.41 hours as of 06.03.2026
Repair/ reclamation History	Reclaimed once at BHSL; between October 2017 and November 2019 after 37,149.34 running hours

The unit-I runner 5020 was in operation from 2017 and its running hours since the commission has reached 30362.41 hours by the time the unit was taken under shutdown for annual maintenance on 30.03.2026.



Figure-2: Erosion observed on runner buckets

The physical condition of the runner buckets was visually inspected, and erosion was observed on all buckets. Considering the significant extent of erosion and it was under took extensive repaired wherever there is major damaged. that the runner had only achieved 30,362.41 hours already operated for more than 71,000 hours, it was decommissioned and replaced with reclaimed runner 2020.

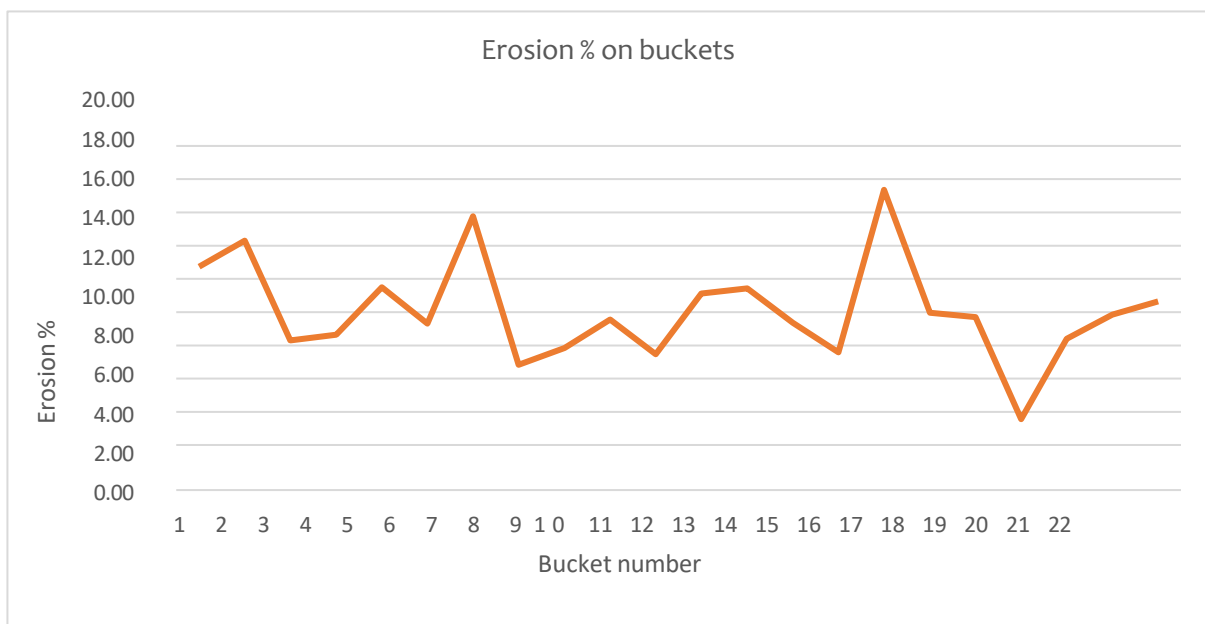


Figure 3: Erosion rates

1. Governor and MIV System

Table 4: Spares consumed for Governor and MIV system

SI #	Description	Quantity	Unit
1	Filter element (5417)	2	No
2	Pressure Switch EDS 3448-5-0250-000 (2-20125)	1	No
3	Cloths	10	M
4	Mulmul Cloths	2	M
5	Banayan cloths	2	M
6	Sandering wheel 5x80	3	No
7	Loctite 406	1	No
8	Cloths	10	M

3.1 Governing System

The governor system is primarily designed to regulate and maintain the turbine rotational speed within permissible limits during no-load operating conditions. This is achieved through continuous sensing of speed variations and rapid modulation of water flow into the turbine, thereby maintaining speed stability in synchronization with the grid frequency reference.

For this generating unit, the SCADA integration works were successfully completed during the previous Annual Maintenance period. Consequently, no major activities associated with the HMC or NCC systems were required during the current maintenance cycle, unlike previous years where these constituted a significant portion of the maintenance scope. The present maintenance activities were therefore limited to routine servicing and reliability assurance works. These included comprehensive cleaning of the OPU tank, inspection and servicing of pump assemblies and oil filtration systems, and restoration of nitrogen pre-charge pressure in the accumulator bank to ensure efficient hydraulic performance. In addition, all high-pressure pipelines connected to the VTA (Valve Table Assembly) were thoroughly inspected for leakage indications, and necessary tightening and rectification works were carried out wherever abnormalities were observed.

3.1.1 Valve Table Assembly (VTA)

The Valve Table Assembly (VTA) forms an integral part of the hydraulic governing system and is primarily designed to regulate and distribute pressurized oil flow to the needle and deflector servomotors. The assembly comprises solenoid-operated control valves, pressure gauges, and associated hydraulic control devices interfaced with the SCADA system to facilitate centralized supervision, remote operation, and real-time monitoring.

Following the successful implementation of the SCADA system during the previous Annual Maintenance, the VTA has been operating with enhanced precision and improved control efficiency over the turbine hydro-mechanical mechanisms. The integration has considerably minimized dependency on manual operation, enhanced system response during operational transients, and improved overall reliability, monitoring capability, and operational stability of the governing system.



Figure 5: Valve Table Assembly

Deflector gap setting was done and the set result is tabulated below.

Table 5: Deflector gap setting readings

Nozzle	Initial Reading (mm)	Final Reading (mm)
J1	16	16
J2	10	8
J3	5.7	8
J4	7.7	8
J5	15.5	8

3.1 MIV System

The Main Inlet Valve (MIV) system functions as the primary turbine isolation arrangement and constitutes the foremost protective barrier of the powerhouse against penstock water ingress and related hydraulic contingencies.

Calibration of the pressure gauges installed in the MIV control panel and pressure accumulator circuit was carried out at HRDC. During the calibration process, four gauges were identified with pressure deviations exceeding the allowable tolerance limits. These instruments shall be replaced upon receipt of newly calibrated gauges. The detailed calibration report is attached for reference.

Considering the operational significance of the hydraulic control pipelines in maintaining the safety, reliability, and functional integrity of the MIV system, periodic Non-Destructive Testing (NDT) is strongly recommended on an annual basis to verify the condition and serviceability of these components under operating conditions.

Accordingly, ultrasonic thickness measurements of the MIV bypass pipeline, decompression pipeline, and pressure accumulators were performed by the HRDC inspection team to evaluate wall thickness reduction, material degradation, and any indication of crack initiation or propagation prior to the development of critical defects. The corresponding inspection and test reports are enclosed herewith.

Furthermore, the servo valve associated with the decompression pipeline was dismantled and internally inspected to identify any accumulation of debris, sludge, or foreign particles. This maintenance activity is recommended during every Annual Maintenance (AM) outage as well as during runner inspection periods to ensure reliable and obstruction-free valve operation. Presence of contaminants within the valve assembly may adversely affect decompression system performance and can lead to abnormal hydraulic pressure fluctuations during operation.

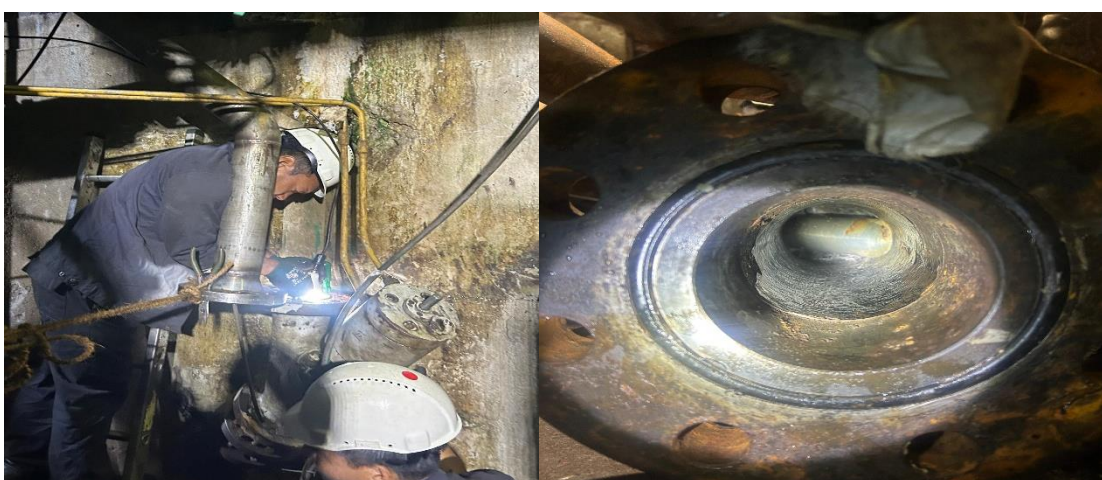


Figure 6: Inspection on MIV decompression line

The OPU pump capacity was checked and verified to be within the normal operating range. Functional checks for internal leakages and servicing were carried out on all directional control solenoid valves installed in the OPU and MIV control panels. The contactors and limit switches of the MIV system were inspected for operational integrity, and replacements were carried out wherever necessary. During the functional testing, no leakage of N_2 gas was observed.

1. Cooling Water System

1.1. Clean Cooling Water (CCW) System

During the Annual Maintenance, both the Centrifugal pumps or Clean Cooling Water Pumps (CCW-pump) was subjected to comprehensive overhauling activities, including complete dismantling, detailed internal inspection, cleaning, and replacement of critical wear components such as mechanical seals, bearings, and gland assemblies.

Subsequent to overhaul, servicing, and reassembly, precision shaft alignment between the pump and drive motor was performed using the Easy-Laser alignment system. The alignment process enabled accurate correction of both angular and parallel misalignments, thereby reducing coupling stress, vibration levels, and undue mechanical loading on rotating components. Final alignment measurements were verified and confirmed to be within the prescribed tolerance limits. Additionally, alignment verification of both the pumps was also carried out.



Figure 7: Overhauling and servicing of CCW pump

1.2 Raw Cooling Water (RCW) System

The pumps were subjected to a detailed visual inspection as part of the preventive maintenance activity. During the inspection, the bearing housings and pump casings were thoroughly cleaned to remove accumulated debris, sludge, and other contaminants.

For these Pumps, the lubricating oil tank was cleaned internally, and the deteriorated oil hose was replaced to ensure reliable lubrication performance. In addition, the Non-Return Valve (NRV) associated with the pump system was replaced due to operational wear.

Furthermore, the motorized strainer and cyclone separator assemblies were dismantled and comprehensively cleaned to restore effective filtration efficiency and maintain uninterrupted fluid flow within the system.



Figure 8: Cleaning of strainer filter elements

These maintenance activities were carried out as preventive measures to maintain system performance and reliability.



Figure 9: Overhauling of RCW pump

Prior to the commencement of the Annual Maintenance (AM) of the generating units, the Hydro Research Development Centre carried out comprehensive condition monitoring and performance assessment of all centrifugal and vertical turbine pumps during unit spinning operation. In addition to vibration monitoring, the HRDC team also performed various diagnostic analyses and operational assessments to evaluate the overall health and performance condition of the rotating equipment.

During the inspection, all operating parameters were thoroughly checked, with particular attention given to abnormal vibration, unusual noise, temperature variation, and overall operational stability of the pumps and associated equipment.

In coordination with the maintenance team, officials from Hydro Research Development Centre subsequently conducted detailed vibration analysis and testing to accurately identify the exact source of vibration, whether originating from the motor, pump assembly, coupling arrangement, or associated rotating components.

Based on the feedback, analysis reports, observations, and technical recommendations provided by the HRDC officials, the annual maintenance and overhauling works of the pumps were executed strictly in accordance with their technical directives and maintenance recommendations to ensure reliable and efficient operation of the equipment.

1.1. Heat Exchanger

Each unit is provided with a shell-and-tube type Main Heat Exchanger (MHE) for supplying clean cooling water to the stator air coolers, OPU coolers, and bearing cooling circuits. Raw Cooling Water (RCW) circulates through the heat exchanger tubes, enabling effective thermal exchange and maintaining the required operating temperature of auxiliary equipment.

As part of the Annual Maintenance (AM) program, the MHE was isolated, dismantled, and subjected to detailed internal inspection. The heat exchanger tubes were mechanically cleaned and thoroughly flushed to remove accumulated scale, sludge, and fouling deposits that could reduce heat transfer efficiency and restrict cooling water flow.

Following completion of the cleaning activities, the end covers were reinstalled and hydrostatic pressure testing was conducted to verify the mechanical integrity and leak tightness of the heat exchanger assembly. During testing, tube leakage was detected in 20 tubes on the inlet side and 16 tubes on the outlet side. The affected tubes were plugged to isolate the defective sections and restore the operational reliability and safety of the cooling water system pending

permanent rectification or tube replacement.

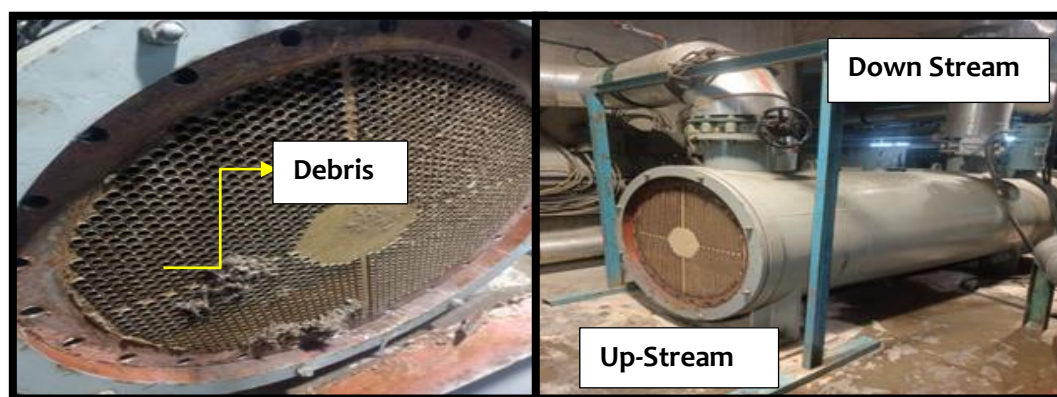


Figure 10: Cleaning of MHE

Table 2: Spares consumed for Cooling system

Sl#	Materials	Quantity	Unit
1	Drive End Mechanical Seal with Gland	1	NO
2	Wearing Ring-CF8M, Part-no. 190	1	NO
3	Bearing-SKF-6039-2z/C3	1	NO
4	SS Ballvalve-50NB with Electric Actuator	1	NO

2. Internal/ external services availed

- Runner inspection (Magnetic Particle Inspection) by HRDC
- Ultrasonic testing on decompression, bypass pipelines, pressure accumulators and distributor pipeline joints by HRDC
- Vibration analysis of CCW and RCW pumps by HRDC
- Calibration of pressure gauges on MIV panels by HRDC
- Testing of Governor and MIV system oil samples by HRDC

1. Maintenance Challenges

During the Annual Maintenance shutdown, detailed inspection of the turbine runner revealed localized cavitation erosion, surface pitting, and material degradation on the runner blades and crown portions. To restore the original hydraulic contour and sustain operational efficiency, repair welding activities were undertaken on the affected areas of the runner assembly. Prior to commencement of welding, all damaged surfaces were thoroughly cleaned, dressed, and prepared in accordance with standard repair procedures. Appropriate welding consumables compatible with the runner base material were utilized as per approved repair specifications and maintenance standards. Upon completion of welding, the repaired zones were subjected to grinding and surface finishing to achieve the required dimensional profile, surface smoothness, and hydraulic contour.

Post-repair inspection confirmed substantial improvement in the overall condition of the turbine runner, including restoration of blade geometry, elimination of surface irregularities, and reduction of cavitation-affected defects. The completed repair welding is expected to enhance hydraulic performance, improve operational reliability, minimize vibration induced by uneven runner surfaces, and extend the service life of the runner during subsequent operation cycles.

Annexure:

S.NO	Annexure	Description
1	Annexure 1	Auxiliary Final Checklist
2	Annexure 2	Deflector mechanical gap setting
3	Annexure 3	Deflector servo timing set
4	Annexure 4	Fasteners tightness checklist
5	Annexure 5	G100 Governor system functionality check
6	Annexure 6	Governor system final checks
7	Annexure 7	Governor nitrogen pressure retention check
8	Annexure 8	Governor OPU pump capacity check
9	Annexure 9	Governor OPU safety valve setting
10	Annexure 10	Governor system pre-checklist
11	Annexure 11	Governor system pre-checklist
12	Annexure 12	Governor system pressure retention check
13	Annexure 13	Inspection of Decompression valve
14	Annexure 14	MIV system final checks
15	Annexure 15	MIV OPU safety valve setting
16	Annexure 16	MIV pre-checklist
17	Annexure 17	MIV system pressure retention checklist
18	Annexure 18	MIV system timing checklist
19	Annexure 19	Nozzle assembly checklist
20	Annexure 20	Nozzle post assembly checklist
21	Annexure 21	Nozzle pre-checklist
22	Annexure 22	Nozzle timing checklist
23	Annexure 23	Turbine Final checklist
24	Annexure 24	Underwater components Inspection
25	Annexure 25	UT report on MIV high-pressure pipelines
26	Annexure 26	UT report on pressure accumulators
27	Annexure 27	Pressure gauge calibration report
28	Annexure 28	Pump Vibration assessment report

Submitted by:

A handwritten signature in blue ink, appearing to be 'Karma', with a long horizontal flourish extending to the right.

(Karma) AEE
Mechanical Maintenance Unit
Tala Hydropower Plant

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